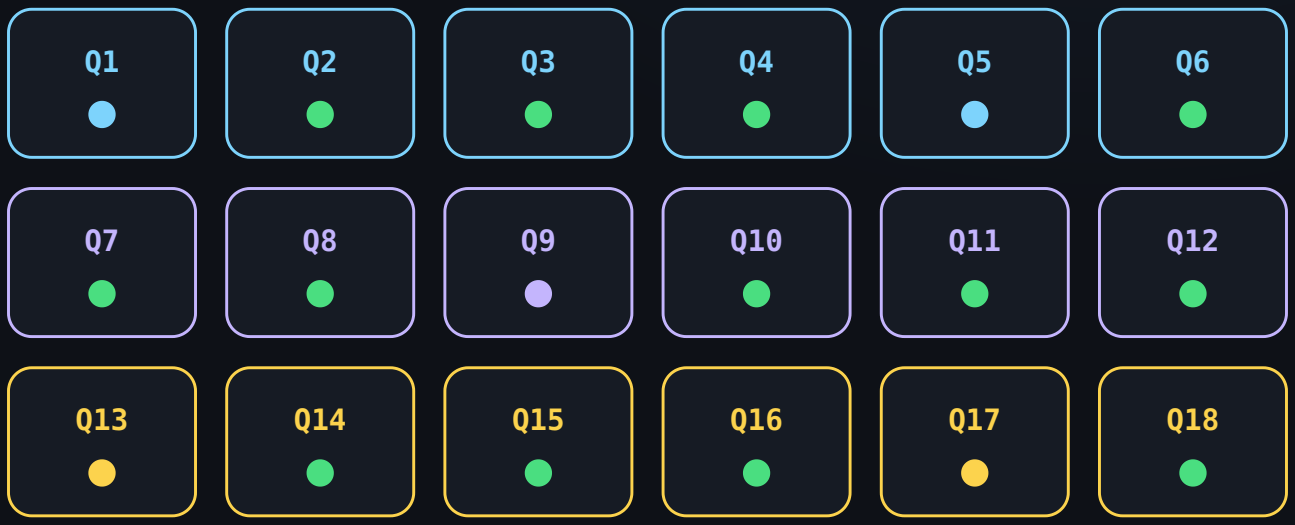
CMPN322 · MOODLE QUIZZES · QUIZ BANK

Computer Graphics Quiz Bank

Your Moodle quiz questions in one place — every question with its options, the correct answer, the working or the lecture sentence behind it, and the file and page it comes from.

Quiz 1 · 24 questions | Quiz 2 · 26 questions | Quiz 3 · 23 questions | Answer key



3	73	13	1
Quizzes	Questions	Reference files	Answer key

Contents

Answer first, then check the green choice and the source

01 **Quiz 1 — Basics, Raster & Color**
24 questions · models, vector vs raster, frame buffer, LUT, direct coding, pipeline
Lec 1 · Lec 2 · SEC 1–3

02 **Quiz 2 — Viewing & Line Drawing**
26 questions · Bresenham, DDA, glViewport, gluOrtho2D, window → viewport, aliasing
Lec 3 · Lec 4 · Lec 5

03 **Quiz 3 — Circle, Ellipse, Transformations & Antialiasing**
23 questions · midpoint circle and ellipse, composite transformations, supersampling
Lec 4 · Lec 5 · Lec 6

04 **Answer Key**
All answers in one table for quick checking
–

How to read each card

- The correct choice is green with ✔ — it is the answer marked correct by Moodle.
- The grey box gives the working (for problems) or the lecture sentence (for theory).
- Top right = the reference file and page. When the quiz uses words that are not in the lecture, the card says so.
- A red ⚠ box marks a quiz answer that does not fully match the lecture method.

غطّي الإجابة الخضرا وحل الأول، وبعدين قارن.

CG / QUIZ BANK / QUIZ 1

Quiz 1

01

Basics of computer graphics, raster images and color

24	Lec 1 · Lec 2	1 mark
questions	main lectures	each

Q1

Source: Lec 1 · p9

------: describes how objects appears (it defines attributes that describe color texture or shape of the object)

a. mathematical model

b. physical model

c. Geometric model ✔

d. all mentioned

Answer: c. Geometric model
Geometric model: describes how objects appears · Mathematical model: describes how object moves.

Q2

Source: Lec 2 · p36–39 · Graphics_SEC 3 · p2

Using the Look-Up Tables (LUTs) method for pixel color representation, how much memory is required for a 4-bit/pixel frame buffer of size 512×1024 along with LUT of 6 byte wide?

a. 1M byte

b. 262144 byte

c. 262240 byte ✔

d. 4K byte

Answer: c. 262240 byte
Frame buffer = 512 × 1024 × 4 / 8 = 262,144 byte · LUT = 2^4 entries × 6 byte = 96 byte · Total = 262,144 + 96 = 262,240 byte

Q3

Source: Lec 1 · p15–16

Which of the following is a characteristic of vector graphics?

a. Always larger in file size than raster images

b. Made up of paths defined by mathematical equations ✔

c. Composed of pixels

d. Limited to a fixed resolution

Answer: b. Made up of paths defined by mathematical equations
Vector: composed of connected sequence of points (paths), a series of mathematical curves · small file size.

Q4 Source: Lec 1 · p15

Which of the following is a primary advantage of using scalable vector graphics (SVG)?

a. It is limited to black and white graphics.

b. It cannot be manipulated with CSS or JavaScript.

c. It only supports raster images.

d. It is resolution-independent and scales without loss of quality. ✔

Answer: d. It is resolution-independent and scales without loss of quality.
Vector graphics print precisely even when they are enlarged · known as scalable vector graphics (SVG).

Q5 Source: Lec 1 · p11

Which of the following is an example of a non-interactive graphics system?

a. Virtual reality simulation

b. CAD software for 3D modeling

c. Video game

d. Animated movie ✔

Answer: d. Animated movie
Non-interactive: the user does not have any control over the image as shown in animation movies.

Q6 Source: Lec 1 · p6 · SEC 1 and 2 · p3

------ analyze pictures to derive descriptions

a. computer graphics

b. computer Vision

c. Image Processing ✔

d. all

Answer: c. Image Processing
Image Processing: analyze pictures to derive descriptions.

Q7 Source: Lec 1 · p16

------ Used for less details Images such as logos

a. Vector display ✔

b. none

c. Raster display

Answer: a. Vector display
Vector: scalable shapes and solid colors (logos and icons) · Raster: digital photos.

Q8

Source: Lec 1 · p11 (the words "real-time rendering" are not in the lecture)

What is a common characteristic of interactive graphics?

a. Static images displayed sequentially

b. Real-time rendering and user feedback ✔

c. Use of traditional print media

d. Pre-rendered animations

Answer: b. Real-time rendering and user feedback
Interactive: the user can make any change with an input device · two-way communication (games).

Q9

Source: Lec 1 · p15

A display that consists of a regular grid of pixels and is normally driven from a frame buffer is called:

a. LED Display.

b. LCD Display.

c. Raster Display. ✔

d. Vector Display.

Answer: c. Raster Display.
Raster displays use a frame buffer to store the color value for each pixel.

Q10

Source: Lec 2 · p19

Bitmap is:

a. Grid of pixels. ✔

b. Grid of points.

c. Grid of edges.

d. Grid of lines.

Answer: a. Grid of pixels.
The array of pixels as a whole is often called a "pixel map" or "bitmap".

Q11

Source: Lec 2 · p26

Can we find a Raster Display in the real life?

a. False.

b. True. ✔

Answer: b. True.
Most displays used today for computer graphics are raster displays (flat panels).

Q12

Source: Graphics_SEC 3 · p1 · Lec 2 · p12

Concave Shape is:

a. Polygon Shape determined if a straight line can be drawn between two points in the shape that lies outside the boundary of the shape. ✔

b. Polygon Shape which have number of edges equal to number of angles.

c. Polygon Shape determined if a straight line can be drawn between two points in the shape that lies inside the boundary of the shape.

d. Shape that have start point and end point not connected.

Answer: a. Polygon Shape determined if a straight line can be drawn between two points in the shape that lies outside the boundary of the shape.
Concave polygon has at least one diagonal lying outside the polygon (curved inward).

Q13

Source: Lec 2 · p10

Elements of Pictures Created in Computer Graphics

a. raster Image

b. Text

c. all of the above ✔

d. polyline

Answer: c. all of the above
Elements of pictures: polyline, text, filled regions, raster images.

Q14

Source: Lec 1 · p12 (the word "RAM" is not in the lecture)

Frame Buffer is:

a. A portion of read-only-memory (ROM) containing a bitmap that drives a video display.

b. A portion of random-access-memory (RAM) containing a bitmap that drives a video display. ✔

c. A portion of Hard-disk containing a bitmap that drives a video display.

d. A portion of flash-memory-register containing a bitmap that drives a video display.

Answer: b. A portion of random-access-memory (RAM) containing a bitmap that drives a video display.
Frame buffer: a block of memory dedicated to graphics output which holds data representing all the pixels in a complete video frame.

Q15

Source: Lec 1 · p24 · SEC 1 and 2 · p3

Graphics pipeline defined as: It is a ..... that describes what steps a graphics system needs to perform to render a ..... to a .....

a. conceptual model - 3D scene - 2D screen ✔

b. Mathematical model - 3D scene - 2D screen

c. Design model - 3D scene - 2D screen

d. conceptual model - 2D scene - 3D screen

Answer: a. conceptual model - 3D scene - 2D screen
Graphics pipeline is a conceptual model that describes what steps a graphics system needs to perform to render a 3D scene to a 2D screen.

Q16 Source: Lec 2 · p28

If a video is recorded at 60 FPS, but played back on a 30 Hz monitor, how many frames will be shown per second?

a. 120 FPS
b. 60 FPS
c. 30 FPS ✔
d. 15 FPS

Answer: c. 30 FPS
Refresh rate (Hz) = how many times per second the monitor can physically update the image → max 30 frames per second.

Q17 Source: Lec 2 · p39

If we use 5 bits pixel value in a 24-bit lookup table representation, how much is the size of the lookup table in bytes?

a. 120 byte
b. 32 byte
c. 96 byte ✔
d. 768 byte

Answer: c. 96 byte
Entries = 2^5 = 32 · size = (2^5 × 24) / 8 = 96 byte

Q18 Source: Lec 2 · p39

If we use 8 bits pixel value in a 15-bit lookup table representation, how many entries does the lookup tables have?

a. 512 entry
b. 15 entry
c. 8 entry
d. 256 entry ✔

Answer: d. 256 entry
Entries = 2^8 = 256 entry (the LUT width does not change the number of entries)

Q19 Source: Graphics_SEC 3 · p2

If we use direct coding of RGB value with 5 bits per primary color, how many possible colors do we have for each pixel?

a. 32 COLOR
b. 32K COLOR ✔
c. 1024 COLOR
d. 512 COLOR

Answer: b. 32K COLOR
$2^5 \times 2^5 \times 2^5 = 32 \times 32 \times 32 = 32{,}768$ colors ≈ 32K

Q20

Source: Lec 2 · p30, p35

In Display, the red, green, and blue components of a pixel each use a pair of bits. These pairs are fed to three digital-to-analog converters (DAC's), which

a. convert logical values into actual voltages ✔

b. converts analog to digital signal

c. choose the color

Answer: a. convert logical values into actual voltages
The converter takes a pixel value and converts it to the corresponding quantity · each DAC has 2 input bits (2^2 = 4 levels).

Q21

Source: Lec 2 · p36 · Graphics_SEC 3 · p1

Pixel color representation can be approached using two common methods: direct coding or Look-Up Tables (LUTs). Which method requires more memory usage?

a. Direct coding. ✔

b. Look-Up Tables (LUTs).

Answer: a. Direct coding.
LUT is a compromise between our desire to have a lower storage requirement and our need to support enough colors → direct coding needs more memory.

Q22

Source: Lec 2 · p12

POLYLINES is:

a. connected sequence of angles.

b. connected sequence of shapes.

c. connected sequence of straight lines. ✔

Answer: c. connected sequence of straight lines.
Polyline: connected sequence of straight lines.

Q23

Source: Lec 1 · p21

The instruction glutInitWindowPosition(100, 150); will apply x, and y coordinate from the origin pixel of display from:

a. left - up. ✔

b. right - up

c. right - down.

d. left - down.

Answer: a. left - up.
glutInitWindowPosition gives an initial location for the upper left corner of the display window.

Q24

Source: Graphics_SEC 3 · p2

Using the direct coding method for pixel color representation, how much memory is required for a 4-bit/pixel frame buffer of size 512×1024?

a. 256K byte ✔

b. 4M byte

c. 2M byte

d. 512K byte

Answer: a. 256K byte

512 × 1024 × 4 / 8 = 262,144 byte = 256K byte

CG / QUIZ BANK / QUIZ 2

Quiz 2

02

Window → viewport, DDA, Bresenham and antialiasing

26	25 / 26	Lec 3 · 4 · 5
questions	your mark	main lectures

Q1

Source: Lec 4 · p30

What is the initial value of the decision parameter (P) when drawing a line using Bresenham's algorithm with endpoints (20,10) and (30,18)?

a. P = 0

b. P = 6 ✔

c. P = 4

d. P = 16

Answer: b. P = 6
Δx = 10, Δy = 8 · p0 = 2Δy − Δx = 16 − 10 = 6

Q2

Source: Lec 3 · p18, p25

Which function sets the viewport in OpenGL?

a. glMatrixMode()

b. glLoadIdentity()

c. glViewport() ✔

d. glOrtho2D()

Answer: c. glViewport()
glViewport(x, y, width, height) sets where the scene is drawn on the screen.

Q3

Source: Lec 4 · p19–22

Which of the following equations is used to calculate the first pixel position between endpoints (7,8) and (3,2) in DDA?

a. (x−1, y−m)

b. (x+1, y+m)

c. (x−(1/m), y−1) ✔

d. (x+(1/m), y+1)

e. (x+1, y−1)

Answer: c. (x−(1/m), y−1)
Δx = −4, Δy = −6 → |m| > 1 → step in y; the line goes down → y − 1, x − 1/m

Q4 Source: Lec 4 · p28

Which of the following equations is used to update the decision parameter in Bresenham's algorithm if the decision parameter value was greater than zero?

a. P_new = P + 2Δy − 2Δx ✔
b. P_new = P + Δx − Δy
c. P_new = P + 2Δy
d. P_new = P + 2Δx

Answer: a. P_new = P + 2Δy − 2Δx
$p_k \geq 0 \rightarrow$ next pixel (x + 1, y + 1), $p_{k+1} = p_k + 2\Delta y - 2\Delta x$

Q5 Source: Lec 4 · p28

Which of the following equations is used to update the decision parameter in Bresenham's algorithm if the decision parameter value is less than zero?

a. P_new = P + 2Δy ✔
b. P_new = P + Δx − Δy
c. P_new = P + 2Δy − 2Δx
d. P_new = P + 2Δx

Answer: a. P_new = P + 2Δy
$p_k < 0 \rightarrow$ next pixel (x + 1, y), $p_{k+1} = p_k + 2\Delta y$

Q6 Source: Graphics sec5 · p1

Which of the following is an advantage of the DDA over other line drawing algorithms?

a. It is simpler to implement ✔
b. It provides better quality lines
c. It uses integer arithmetic exclusively
d. It requires less memory

Answer: a. It is simpler to implement
DDA advantages: simplicity, speed relative to direct equation evaluation.

Q7 Source: Lec 1 · p21 · Lec 3 · p16

Which OpenGL function is used to set the 2D world window?

a. glViewport()
b. gluOrtho2D() ✔
c. glMatrixMode()
d. glLoadIdentity()

Answer: b. gluOrtho2D()
gluOrtho2D(xmin, xmax, ymin, ymax) sets the world window (clipping region).

Q8

Source: Lec 3 · p5–6

Consider the window and viewport. The window has (W.l, W.r, W.b, W.t) = (50, 110, 70, 130) and the viewport has (V.l, V.r, V.b, V.t) = (20, 80, 20, 80). For the point (XW, YW) = (60, 90), what are the corresponding viewport coordinates (XV, YV)?

a. (35, 40)

b. (40, 60)

c. (30, 40) ✔

d. (35, 60)

Answer: c. (30, 40)
Sx = (80 − 20)/(110 − 50) = 1 · Sy = (80 − 20)/(130 − 70) = 1 · XV = 20 + 1(60 − 50) = 30 · YV = 20 + 1(90 − 70) = 40

Q9

Source: Lec 4 · p19–23

DDA Line Drawing Algorithm produce the following points for a line with endpoints (5,7) and (8,14).

a. (5,9) (6,10) (6,11)

b. (5,7) (5,8) (6,10)

c. (7,11) (7,12) (7,13) ✔

Answer: c. (7,11) (7,12) (7,13)
Δx = 3, Δy = 7 → |m| > 1, 1/m = 0.43, y + 1, x + 0.43: (5.43, 8)→(5,8) · (5.86, 9)→(6,9) · (6.29, 10)→(6,10) · (6.71, 11)→(7,11) · (7.14, 12)→(7,12) · (7.57, 13)→(8,13) · (8,14)

⚠ Quiz answer = C. With the lecture rounding (int(a + 0.5)) the point at y = 13 is (8,13), not (7,13); (7,11) and (7,12) are correct. A and B are wrong (B has the start point and (5,8),(6,10) skip y = 9).

Q10

Source: Lec 4 · p24 · Graphics sec5 · p1

From the disadvantage of DDA

a. 1 and 2 ✔

b. 1_The floating-point arithmetic is time-consuming

c. 2_The accumulation of round-off error in successive additions of the floating-point increment.

d. 2 and 3

e. 3_We cannot use the multiplication method.

Answer: a. 1 and 2
DDA: floating-point arithmetic is time-consuming · accumulation of round-off error makes pixel positions drift from the true line path.

Q11

Source: Lec 5 · p24

However, the jaggies (stair step appearance) will be apparent to some extent even at the highest resolution possible with current technology.

a. false

b. true ✔

Answer: b. true
Even at the highest resolution possible with current technology, the jaggies will be apparent to some extent.

Q12

Source: Lec 4 · p26–28

If the slope of line is small negative, we decide to change or not each time the value of ------

a. Y value ✔

b. X value

c. non

d. decide each time step which to increment

Answer: a. Y value
|m| < 1 → x steps by 1 every time, and the decision parameter decides if y changes.

Q13

Source: Lec 3 · p5

If the world window has coordinates (0, 2, 0, 1) and the viewport has coordinates (40, 400, 60, 300), what is the scaling factor Sx?

a. 0.5

b. 180 ✔

c. 240

d. 2.0

Answer: b. 180
Sx = (V.r − V.l)/(W.r − W.l) = (400 − 40)/(2 − 0) = 180

Q14

Source: Lec 4 · p27–28

In Bresenham's algorithm, the decision parameter (P) is used to determine:

a. The color of the line

b. The slope of the line

c. Whether to increment y or not ✔

d. The length of the line

Answer: c. Whether to increment y or not
For $x_k + 1$ we choose $(x_k + 1, y_k)$ or $(x_k + 1, y_k + 1)$ — the sign of P decides.

Q15

Source: Lec 4 · p28

In Bresenham's algorithm, when the decision parameter (P) is negative, the next pixel:

a. Is diagonally down and to the right

b. Is diagonally up and to the right

c. Is directly to the right ✔

d. Is directly above

Answer: c. Is directly to the right
$p_k < 0$ → next point $(x_k + 1, y_k)$: same y, one step right.

Q16 Source: Lec 3 · p3 · Graphics_Sec 4 · p1

In the context of window-to-viewport mapping, what does the "window" represent?

a. The clipping region in device coordinates

b. The normalized coordinate system

c. The area on the screen where the image is displayed

d. The region of the world that is to be displayed ✔

Answer: d. The region of the world that is to be displayed
Window: what is to be displayed (world coordinates) · Viewport: where it is to be displayed.

Q17 Source: Lec 4 · p11, p19

In the DDA algorithm, what is the significance of the variable m?

a. It represents the midpoint of a line segment

b. It is used to calculate the length of the line

c. It determines the color of the line

d. It indicates the slope of the line ✔

Answer: d. It indicates the slope of the line
$y = m \cdot x + b$, with m as the slope of the line · DDA adds m (or 1/m) each step.

Q18 Source: Lec 4 · p28

In which algorithm is the decision parameter updated during the line drawing process?

a. Neither algorithm

b. Bresenham's Midpoint Algorithm ✔

c. Both algorithms

d. Digital Differential Analyzer (DDA)

Answer: b. Bresenham's Midpoint Algorithm
Bresenham: $p_{k+1} = p_k + 2\Delta y$ or $p_k + 2\Delta y - 2\Delta x$ · DDA has no decision parameter.

Q19 Source: Lec 4 · p24–25

The main advantage of Bresenham's algorithm over DDA is:

a. It produces higher-quality lines

b. It can handle all types of lines equally well

c. It requires less computational overhead ✔

d. It is easier to implement

Answer: c. It requires less computational overhead
Bresenham: accurate and efficient, uses only incremental integer calculations · DDA: floating-point arithmetic is time-consuming.

Q20

Source: Lec 4 · p25

The primary advantage of Bresenham's algorithm over other line drawing algorithms is:

a. It is faster and uses only integer calculations ✔

b. It can draw curves easily

c. It requires more memory

d. It uses floating-point arithmetic

Answer: a. It is faster and uses only integer calculations
Bresenham's line algorithm: accurate and efficient · uses only incremental integer calculations.

Q21

Source: Lec 3 · p8

What is the default viewport in OpenGL if the screen window has width W and height H?

a. (W, H, 0, 0)

b. (−1, −1, 1, 1)

c. (0, 0, W, H) ✔

d. (0, 0, 1, 1)

Answer: c. (0, 0, W, H)
By default the viewport is the entire screen window: lower left corner at (0, 0) and upper right corner at (W, H).

Q22

Source: Lec 3 · p2

What is the main goal of using normalized coordinates in graphics systems?

a. To ensure device independence ✔

b. To enable 3D transformations

c. To improve rendering speed

d. To simplify mathematical computations

Answer: a. To ensure device independence
To ensure device independence, graphics systems convert object descriptions to normalized coordinates.

Q23

Source: Lec 4 · p2

What is the primary purpose of the Digital Differential Analyzer?

a. To store large datasets

b. To perform arithmetic operations

c. To manage memory allocation

d. To generate and display curves and lines ✔

Answer: d. To generate and display curves and lines
Popular algorithms used to generate lines are DDA and Bresenham's midpoint algorithm.

Q24

Source: Graphics_Sec 4 · p1

What is the primary purpose of the window-to-viewport transformation?

a. To normalize object colors

b. To convert 3D coordinates to 2D coordinates

c. To rotate and scale objects in the world window

d. To map a world-coordinate scene to device coordinates ✔

Answer: d. To map a world-coordinate scene to device coordinates
Window-to-viewport mapping: a scene defined in world coordinates (window) is mapped to screen coordinates (viewport).

Q25

Source: Lec 3 · p10 (code only — the role is not written in the slides)

What is the role of glLoadIdentity() in the OpenGL initialization code?

a. To reset the current matrix to the identity matrix ✔

b. To load the vertex data

c. To define the viewport size

d. To set the background color

Answer: a. To reset the current matrix to the identity matrix
In myInit: glMatrixMode(GL_PROJECTION); glLoadIdentity(); → the projection matrix starts as identity before gluOrtho2D.

Q26

Source: Lec 3 · p25

When automatically setting the viewport to preserve the aspect ratio, if R > W/H, which of the following is used?

a. glViewport(0, 0, H*R, H)

b. glViewport(0, 0, W, W/R) ✔

c. glViewport(0, 0, W, H)

d. glViewport(0, 0, W/R, H)

Answer: b. glViewport(0, 0, W, W/R)
R > W/H → width W and height W/R → glViewport(0, 0, W, W/R) · R < W/H → glViewport(0, 0, H*R, H)

CG / QUIZ BANK / QUIZ 3

Quiz 3

Circle, ellipse, transformations and antialiasing

03

23	23 / 23	Lec 4 · 5 · 6
questions	your mark	main lectures

Q1

Source: Lec 1 · p7 (the lecture says "physics")

------ to model light and to perform simulations for animation

a. all mentioned
b. mathematical model
c. physical model ✔
d. Geometric model

Answer: c. physical model
We use physics to model light and to perform simulations for animation.

Q2

Source: Lec 6 · p26, p32

Rotate point p by angle 70 degree, then rotate it by angle −20, then scale it to half horizontally. Which of the following transformations correctly produces the point p′?

a. P′ = r(70)·s(0.5,1)·r(−20)·p
b. P′ = s(0.5,1)·r(50)·p ✔
c. P′ = r(70)·r(−20)·s(1,0.5)·p

Answer: b. P′ = s(0.5,1)·r(50)·p
R(−20)·R(70) = R(70 − 20) = R(50) (rotation angles add) · then S(0.5, 1) is written on the left → P′ = S(0.5,1)·R(50)·P

Q3

Source: Lec 5 · p11–13

By using midpoint algorithm to draw a circle with a radius of 9 units, along the circle octant in the 1st quadrant includes

a. (1,9), (2,9), (3,9), (4,9), (5,9), (8,8), (7,7) ✔
b. (1,9), (2,10), (3,11), (4,11), (5,11), (8,12), (7,13)
c. (1,10), (2,9), (3,9), (4,9), (5,9), (8,8), (7,7)

Answer: a. (1,9), (2,9), (3,9), (4,9), (5,9), (8,8), (7,7)
P0 = 1 − 9 = −8 → (1,9) · P = −5 → (2,9) · P = 0 ≥ 0 → (3,8) · P = −7 → (4,8) · P = 2 → (5,7) · P = −3 → (6,7) · P = 10 → (7,6)

⚠ Quiz answer = A. With the lecture method the octant is (0,9) (1,9) (2,9) (3,8) (4,8) (5,7) (6,7) (7,6); only (1,9) and (2,9) of A match. A is the only choice with y never above r = 9, so it is the closest.

Q4

Source: Lec 4 · p24–25

Which algorithm may produce more rounding errors due to floating-point arithmetic?

a. Both algorithms

b. Neither algorithm

c. Bresenham's Midpoint Algorithm

d. Digital Differential Analyzer (DDA) ✔

Answer: d. Digital Differential Analyzer (DDA)
DDA: the accumulation of round-off error in successive additions of the floating-point increment · Bresenham uses only integers.

Q5

Source: Graphics_SEC 3 · p2

If we use direct coding of RGB value with 5 bits per primary color, how many possible colors do we have for each pixel?

a. 32 COLOR

b. 512 COLOR

c. 32K COLOR ✔

d. 1024 COLOR

Answer: c. 32K COLOR
$2^5 \times 2^5 \times 2^5 = 32{,}768$ colors ≈ 32K

Q6

Source: Lec 4 · p28–30

Bresenham line drawing for a line with endpoints (10, 0) and (20, 8). This line has a slope of 0.8, includes

a. (12,2), (13,3), (14,4), (15,4)

b. (12,2), (13,3), (14,3), (15,4)

c. (12,2), (13,2), (14,3), (15,4) ✔

Answer: c. (12,2), (13,2), (14,3), (15,4)
$\Delta x = 10$, $\Delta y = 8$ · p0 = 6 → (11,1) · p = 2 → (12,2) · p = −2 → (13,2) · p = 14 → (14,3) · p = 10 → (15,4)

Q7

Source: Lec 5 · p20

Using the midpoint ellipse drawing algorithm, what's the initial point used to draw the ellipse with rx of 8 units and ry of 10 units in the 1st quadrant?

a. (8,10)

b. (0,8)

c. (0,10) ✔

d. (10,8)

Answer: c. (0,10)
Region 1 initial point (x0, y0) = $(0, r_y) = (0, 10)$

Q8 Source: Lec 4 · p28

What is the initial value of the decision parameter (P) when drawing a line using Bresenham's algorithm?

a. P = 0

b. P = Δx − Δy

c. P = Δy − Δx

d. P = 2Δy − Δx ✔

Answer: d. P = 2Δy − Δx
Obtain the starting value for the decision parameter p0 = 2Δy − Δx.

Q9 Source: Lec 5 · p24

There is no limit to how big we can make the frame buffer and still maintain the refresh rate at 60 frames or more per second.

a. true

b. false ✔

Answer: b. false
There is a limit to how big we can make the frame buffer and still maintain the refresh rate at 60 frames or more per second.

Q10 Source: Graphics_SEC 3 · p1 · Lec 2 · p12

Convex Shape is:

a. Polygon Shape determined if a straight line can be drawn between two points in the shape that lies outside the boundary of the shape.

b. Polygon Shape determined if a straight line can be drawn between two points in the shape that lies inside the boundary of the shape. ✔

c. Polygon Shape which have number of edges equal to number of angles.

d. Shape that have start point and end point not connected.

Answer: b. Polygon Shape determined if a straight line can be drawn between two points in the shape that lies inside the boundary of the shape.
Convex polygon has all its diagonals lying inside the polygon (curved outward).

Q11 Source: Graphics_Sec 4 · p1 · Lec 3 · p5

If the world coordinates of a point are (xW, yW) and the corresponding viewport coordinates are (xV, yV), what is the basic formula to convert from world coordinates to viewport coordinates?

a. XV = XVmin + (XVmax − XVmin)/(XWmax − XWmin)

b. XV = XVmin + ((XVmax − XVmin)/(XWmax − XWmin)) * (XW − XWmin) ✔

c. XV = (XVmax − XVmin)/(XWmax − XWmin) * (XW − XWmin)

d. xv = xw/2

Answer: b. XV = XVmin + ((XVmax − XVmin)/(XWmax − XWmin)) * (XW − XWmin)
XV = XVmin + Sx·(XW − XWmin), Sx = (XVmax − XVmin)/(XWmax − XWmin)

Q12

Source: Graphics_Sec 7 · p3 · Lec 6 · p7

A triangle is defined by three points A(2,2), B(3,5), C(4,4). What's the transformation coordinate after horizontal reflection?

a. A(−2,−2), B(−3,−5), C(−4,−4)

b. A(2,−2), B(3,−5), C(4,−4) ✔

c. A(−2,2), B(−3,5), C(−4,4)

Answer: b. A(2,−2), B(3,−5), C(4,−4)
Horizontal reflection: $x' = x$, $y' = -y \rightarrow A(2,-2), B(3,-5), C(4,-4)$

Q13

Source: Lec 5 · p15

In the Midpoint Ellipse Algorithm, the two axes of the ellipse are defined by which of the following parameters?

a. Radius and Diameter

b. X and Y Coordinates

c. Major and Minor Axes ✔

d. Height and Width

Answer: c. Major and Minor Axes
Ellipse with semimajor axis and semiminor axis (r_x, r_y).

Q14

Source: Lec 2 · p39

If we use 8 bits pixel value in a 15-bit lookup table representation, how many entries does the lookup tables have?

a. 512 entry

b. 15 entry

c. 256 entry ✔

d. 8 entry

Answer: c. 256 entry
Entries = 2^8 = 256 entry

Q15

Source: Lec 5 · p25

Which of the following techniques is commonly used for antialiasing?

a. Supersampling ✔

b. Texture mapping

c. Rasterization

d. Depth buffering

Answer: a. Supersampling
A straightforward antialiasing method … this technique is called super-sampling.

Q16

Source: Lec 5 · p15–16

What are the main parameters needed to implement the Midpoint Ellipse Algorithm?

a. Only the center point

b. Center coordinates and the lengths of the major and minor axes ✔

c. Radius and angle

d. The four corners of the ellipse

Answer: b. Center coordinates and the lengths of the major and minor axes
The ellipse is defined by its center and r_x, r_y (semimajor / semiminor axes).

Q17

Source: Lec 5 · p12 · Graphics_Sec 6 · p1

Using the midpoint circle drawing algorithm, what's the initial point used to draw a circle with a radius of 8 units, along the circle octant in the 1st quadrant?

a. (8,8)

b. (8,0)

c. (0,8) ✔

d. (0,0)

Answer: c. (0,8)
Initial point (0, r) = (0, 8)

Q18

Source: Lec 5 · p20–21

What is the purpose of the decision parameter in the Midpoint Ellipse Algorithm?

a. To calculate the area of the ellipse

b. To decide which pixel to plot next ✔

c. To set the rendering speed

d. To determine the color of the ellipse

Answer: b. To decide which pixel to plot next
If $p_k < 0$ the midpoint is inside the ellipse · if $p_k > 0$ it is outside → choose the next pixel.

Q19

Source: Lec 6 · p20–22, p30

Rotate point p by angle 30 around point (4,3) then rotate by angle −50 around point (2,2). Which of the following transformations correctly produces the point p′?

a. P′ = T(2,2)·r(−50)·T(2,1)·r(30)·T(−4,−3)·p ✔

b. P′ = (30)·T(−4,−3)·T(4,3)·r(−50)·T(−2,−2)·T(2,2)·p

c. P′ = r(30)·T(−4,−3)·r(−50)·T(−2,−2)·p

Answer: a. P′ = T(2,2)·r(−50)·T(2,1)·r(30)·T(−4,−3)·p
First: T(4,3)·R(30)·T(−4,−3) · then: T(2,2)·R(−50)·T(−2,−2) · the middle translations add: T(−2,−2)·T(4,3) = T(2,1)

Q20 Source: Lec 5 · p24

What is the primary purpose of antialiasing in computer graphics?

a. To minimize the visual distortion of edges ✔
b. To enhance texture details
c. To increase rendering speed
d. To reduce color depth

Answer: a. To minimize the visual distortion of edges
The jagged (stair-step) appearance is a distortion from under sampling; antialiasing methods compensate for it.

Q21 Source: Lec 4 · p22

Which of the following equations is used to calculate the first pixel position if the slope of line is greater than 1 and dx and dy are negative values in DDA?

a. (x−1, y−m)
b. (x+1, y+m)
c. (x−(1/m), y−1) ✔
d. (x+1, y−1)
e. (x+(1/m), y+1)

Answer: c. (x−(1/m), y−1)
|m| > 1 → step in y · line goes down and left → y − 1, x − 1/m

Q22 Source: Graphics_Sec 6 · p1

What is the initial decision parameter used in the Midpoint Circle Drawing Algorithm for the first point of the circle?

a. p = r − 2
b. p = 1 − r ✔
c. p = r
d. p = r*r

Answer: b. p = 1 − r
Initial point (0, r) & P0 = 1 − r

Q23 Source: Lec 5 · p10, p16

If the points of an ellipse that lie in the first quad are (1,6), (2,6), (3,6) and (4,5), then the similar points on the second quad are:

a. (−1,6), (−2,6), (−3,6), (−4,5) ✔
b. (1,−6), (2,−6), (3,−6), (4,−5)
c. (−1,−6), (−2,−6), (−3,−6), (−4,−5)

Answer: a. (−1,6), (−2,6), (−3,6), (−4,5)
Symmetry: second quadrant = (−x, y) · third (−x, −y) · fourth (x, −y)

CG / QUIZ BANK / ANSWER KEY

Answer Key

04

All 73 answers on one page

#	QUIZ 1	QUIZ 2	QUIZ 3
Q1	c Geometric model	b P = 6	c physical model
Q2	c 262240 byte	c glViewport()	b P′ = s(0.5,1)·r(50)·p
Q3	b Made up of paths defined…	c (x−(1/m), y−1)	a (1,9), (2,9), (3,9), (4,…
Q4	d It is resolution-indepen…	a P_new = P + 2Δy − 2Δx	d Digital Differential Ana…
Q5	d Animated movie	a P_new = P + 2Δy	c 32K COLOR
Q6	c Image Processing	a It is simpler to impleme…	c (12,2), (13,2), (14,3), …
Q7	a Vector display	b gluOrtho2D()	c (0,10)
Q8	b Real-time rendering and …	c (30, 40)	d P = 2Δy − Δx
Q9	c Raster Display.	c (7,11) (7,12) (7,13)	b false
Q10	a Grid of pixels.	a 1 and 2	b Polygon Shape determined…
Q11	b True.	b true	b XV = XVmin + ((XVmax − X…
Q12	a Polygon Shape determined…	a Y value	b A(2,−2), B(3,−5), C(4,−4…
Q13	c all of the above	b 180	c Major and Minor Axes
Q14	b A portion of random-acce…	c Whether to increment y o…	c 256 entry
Q15	a conceptual model - 3D sc…	c Is directly to the right	a Supersampling
Q16	c 30 FPS	d The region of the world …	b Center coordinates and t…
Q17	c 96 byte	d It indicates the slope o…	c (0,8)
Q18	d 256 entry	b Bresenham's Midpoint Alg…	b To decide which pixel to…
Q19	b 32K COLOR	c It requires less computa…	a P′ = T(2,2)·r(−50)·T(2,1…
Q20	a convert logical values i…	a It is faster and uses on…	a To minimize the visual d…
Q21	a Direct coding.	c (0, 0, W, H)	c (x−(1/m), y−1)
Q22	c connected sequence of st…	a To ensure device indepen…	b p = 1 − r
Q23	a left - up.	d To generate and display …	a (−1,6), (−2,6), (−3,6), …
Q24	a 256K byte	d To map a world-coordinat…	
Q25		a To reset the current mat…	
Q26		b glViewport(0, 0, W, W/R)	